Import library

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Import dataset

```
from sklearn.datasets import load_iris
data = load_iris(as_frame=True)
df = data.frame
print(df.head())
print(df.shape)
print(df['target'].value_counts())
```

```
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2

   target
0       0
1       0
2       0
3       0
4       0
(150, 5)
target
0    50
1    50
2    50
Name: count, dtype: int64
```

Pisahkan fitur (X) dan target (y)

```
X = df.drop('target', axis=1)
y = df['target']
```

Split data latihan dan uji

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)
```

Feature scaling

```
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
```

Latih model Naive Bayes

```
from sklearn.naive_bayes import GaussianNB
model = GaussianNB()
model.fit(X_train, y_train)
```

```
▾ GaussianNB
GaussianNB()
```

Prediksi

```
y_pred = model.predict(X_test)
```

Evaluasi

```
from sklearn.metrics import confusion_matrix, accuracy_score, classification_report

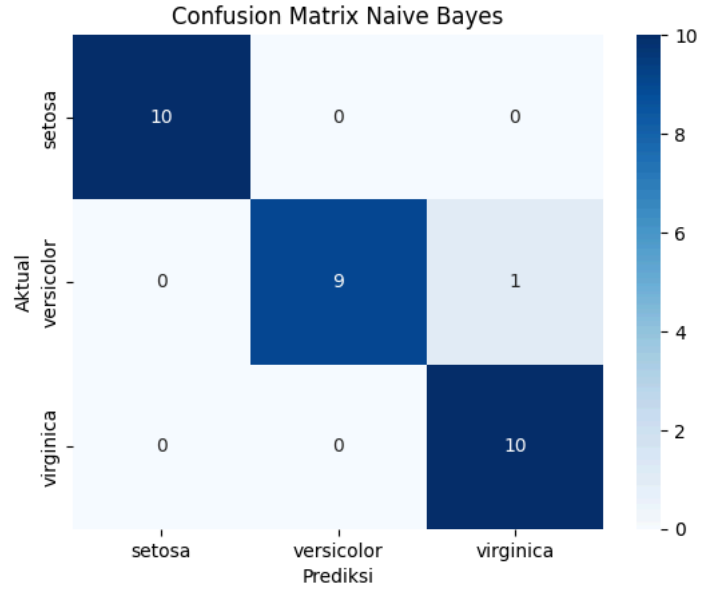
cm = confusion_matrix(y_test, y_pred)
print(cm)
print("Akurasi:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred, target_names=data.target_names))

sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=data.target_names, yticklabels=data.target_names)
plt.xlabel('Prediksi')
plt.ylabel('Aktual')
plt.title('Confusion Matrix Naive Bayes')
plt.show()
```

```
[[10  0  0]
 [ 0  9  1]
 [ 0  0 10]]
Akurasi: 0.9666666666666667
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       1.00      0.90      0.95        10
   virginica       0.91      1.00      0.95        10

    accuracy                           0.97        30
   macro avg       0.97      0.97      0.97        30
weighted avg       0.97      0.97      0.97        30
```



Akurasi model (96,67%). Kelas setosa diprediksi sempurna, sedangkan satu versicolor tertukar dengan virginica. Naive Bayes cukup baik untuk dataset Iris